problem. Let’s use the acronym MPC (Micro Payment Channel) to represent the Dapp.

> **PROBLEM STATEMENT** Assume that a United Nations-like nongovernmental organization wants to pay individuals certain incentive payments (micropayments) for every bin of recyclable plastics collected from the environment and deposited at a designated location for recycling and proper disposal. You are required to design and develop a decentralized solution to facilitate this process.

Here are some assumptions and further details:

- Some verification mechanism exists to make sure that the bins of plastics collected contain the right amount and correct types of plastics. Otherwise, bins are rejected.
- A human worker or even a robot collects plastics in bins and deposits them at designated locations. Every time the bins collected by a worker are verified, a message is sent to the patron organization. On receiving a message, the patron organization sends an authorized off-chain micropayment through a channel established between the organization and the worker. Potentially, there may be many micropayments paid to the worker in a single session of plastics collection. In a given session (such as a day), the value of a micropayment is the sum of all the previous micropayments added to the current one.
- Instead of cashing these small payments every time a bin of plastic garbage is collected and incurring fees, the worker waits until the last bin of the day and then receives the payment through one on-chain Tx. By design, this single Tx request is for the value of the last micropayment because it holds the accumulated value.